

SMIM — Shared Memory Integrity Module

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Independent Methodological Authority

OriginID: OOF-OID-MEM-SMIM-2026-06-08-0001

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Group Memory Governance Layer

Governed Space: Shared Memory Integrity

Category: AI & Interpretation

Subcategory: Shared Memory Governance

Type: Group Memory Integrity Module

Parent Standard: Group Memory Integrity Standard (GMIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 8 June 2026

Compatibility: OOF Methodology OS · Group Memory Integrity Standard (GMIS) ·

Memory Integrity Standard (MIS) · Collective Cognition Integrity Standard (CCIS) ·

Multi-Agent Cognition Integrity Standard (MCIS) · Human-AI Cognition Integrity Standard (HAICS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Shared Memory Integrity Module (SMIM) defines the structural conditions under which memory shared by multiple participants remains coherent, accessible, synchronized, reconstructable, accountable, and operationally valid throughout collaborative activities, coordination processes, collective decision-making, and long-term group operation.

SMIM governs shared memory.

The module ensures that participants operate from a sufficiently aligned memory foundation rather than fragmented, isolated, contradictory, or inaccessible memory states.

A system satisfies SMIM only if:

- shared memory remains accessible
- shared memory remains coherent
- shared memory remains reconstructable
- memory access remains governable
- memory utilization remains accountable
- shared memory remains operationally valid

A collaborative environment that cannot maintain a reliable shared memory layer does not satisfy SMIM.

Module Operational Role

SMIM defines the shared-memory governance layer of GMIS.

Its role is to preserve common memory continuity across participants.

Module Operational Space

SMIM governs:

- shared memory
- collaborative memory
- common knowledge memory
- memory accessibility
- shared context preservation
- memory synchronization
- memory accountability
- shared memory governance

The module applies wherever multiple participants rely on common memory resources.

Module Function

The module applies wherever systems must preserve:

- collective continuity
- shared knowledge
- collaborative coordination
- governance-valid memory access
- reconstructable shared history
- reliable group operation

Its function is to ensure that groups operate from a trustworthy shared memory foundation.

Minimum Implementation Framework

1. Define the Shared Memory Object

The organization must define which memory assets require shared-memory governance.

This may include:

- shared project knowledge
 - operational procedures
 - collaborative history
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- team knowledge
- Human-AI shared memory
- multi-agent memory
- organizational knowledge
- collective operational memory

2. Define Shared Memory Conditions

The system must define the conditions under which shared memory remains valid.
This includes:

- accessibility requirements
- continuity requirements
- synchronization requirements
- reconstruction requirements
- accountability requirements
- governance-valid shared-memory conditions

3. Define Shared Memory Degradation Detection Logic

The system must define how shared-memory degradation is identified.
This may include:

- memory fragmentation
- inaccessible shared knowledge
- contradictory memory states
- synchronization failures
- context divergence
- collaborative memory loss

4. Define Operational Response or Governance Logic

The system must define governance logic for shared-memory failures.
Governance response may include:

- memory review
- synchronization activities
- reconstruction procedures
- governance intervention
- escalation
- memory restoration
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- shared-memory changes
- synchronization events

- governance reviews
- intervention activities
- accountability assignments
- resulting memory states

A group-memory environment must not remain shared-memory-valid if materially significant shared memory cannot be reconstructed, reviewed, synchronized, or governed.

Use Case 1 — Human-AI Operations Team

Scenario

Humans and AI systems collaborate on operational tasks while relying on shared procedures, objectives, and historical knowledge.

Application

SMIM governs preservation of shared memory, continuity of common knowledge, and collaborative accessibility.

Result

The team gains stronger coordination, improved continuity, and reduced exposure to knowledge fragmentation.

Use Case 2 — Multi-Agent Intelligence Environment

Scenario

Multiple autonomous agents continuously exchange information and utilize common operational memory while executing coordinated tasks.

Application

SMIM governs shared memory integrity, synchronization, and accessibility across agent environments.

Result

The environment gains stronger operational coherence, improved collaboration reliability, and reduced exposure to shared-memory failures.

Canonical Closing Statement

Shared Memory Integrity Module (SMIM) defines the structural conditions under which memory shared by multiple participants remains coherent, accessible, synchronized, reconstructable, accountable, and operationally valid throughout collaborative activities, coordination processes, collective decision-making, and long-term group operation.

**Collective continuity cannot exist without trustworthy shared memory.
Shared memory integrity therefore becomes a foundational integrity
condition of collaborative intelligence systems.**

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Canonical Source

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